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Profile

Game Design and Development student at the University of Greenwich focused on gameplay programming and technical design. I build playable Unity and Unreal Engine projects across FPS research tools, VR interaction, AI behavior, UI systems, multiplayer synchronization, Unreal materials, HLSL shader work and Git-based team production.

Education

University of Greenwich | BSc (Hons) Game Design and Development *Expected September 2026*

Technical Skills

Programming: C#, C++, Python, HLSL shader programming
Engines: Unity, Unreal Engine 5, Blueprint, XR Interaction Toolkit, NavMesh, Niagara VFX
Production: Git, GitHub, Agile planning, Scrum-style sprints, playtesting, documentation
Art and UI: Maya, Blender, Unreal materials, post-process effects, mobile UI, world-space VR UI
Languages: Tamil, English, Hindi

Academic and Technical Projects

Repetitive Aim Training FPS Study — *Unity | C#*

Final Year Individual Project | Solo Developer and Researcher

- Designed two linked FPS builds: a Training Lab for controlled aim drills and a Gameplay Environment for combat testing, comparing players who trained first against players who only completed the FPS test.
- Implemented flick, tracking, recoil, target-switching and round-based shooting systems with NavMesh enemies, weapon spread, hit feedback, scoring, time pressure and controlled player progression.
- Captured CSV telemetry for shots, hits, headshots, kills, accuracy, first-hit latency, time-to-kill, recoil error and hits per minute so the final paper could analyse measurable performance changes.

Echoes of The Twin Tombs — *Unity | C#*

VR Gameplay Engineer and Interface Systems Programmer

- Developed cooperative VR escape-room mechanics for Meta Quest 3 using physical interactions, world-space UI, role selection, role-based spawning, reset behaviors and clear in-headset guidance.
- Owned specialist systems across the Astral Relay, Prism Reactor and Lever/Trapdoor puzzle flow, including shared puzzle state, player communication cues and fail-safe logic for progression.
- Worked on networked player synchronization and multiplayer room flow so separated players could solve communication-led puzzles without puzzle states desynchronizing or blocking completion.

LumberJill: The RPG — *Unity | C#*

Lead Programmer and Systems Designer

- Owned core gameplay systems for a mobile workshop RPG, including machine processing, crafting grid validation, storage flow, touch UI, job board tasks and customer delivery loops.
- Integrated real-world lumber price data into a readable in-game economy using controlled fluctuation, trend sensitivity, stock-style UI display and offline progression systems.
- Built sustainability loops around reforestation, seed conversion, timed tree growth and production management so the mechanics supported the project theme instead of acting as decoration.

UnCaged: The Wild After Party — *Unreal Engine 5 | Blueprint*

Solo Developer

- Built a third-person survival prototype with sprint, crouch, stamina, inventory flow, pickup guidance, checkpoints, enemy health UI, basic combat feedback and final escape logic.
- Created forest level structure using Landscape tools, Landmass, spline shaping, fog, lighting, ambient sound, route blocking, rope bridge testing, NavMesh enemies and optimization passes.
- Designed the game loop around exploration, resource collection, stealth pressure, limited combat and a final gate sequence so the prototype could be completed within a short session.

WeatherSys — *Unreal Engine 5 | C++*

C++ Plugin Developer

- Built a runtime procedural weather plugin that reads Open Weather API data and converts live conditions into rain, snow, fog, cloud cover, lighting, sunrise, sunset, night and humidity responses.
- Created a Slate developer panel for assigning scene components, forcing weather states, testing values, controlling update intervals and avoiding heavy Blueprint dependency during setup.

- Designed the system for gradual transitions and editor-safe overrides so designers could test cinematic weather states without waiting for real-time API conditions.

Reactive Containment Chamber *Unreal Engine 5 | HLSL*
Technical Artist and Shader Programmer
- Created a sci-fi shader showcase with reactive liquid, translucent glass, pulsing specimen materials, animated cables, floor circuitry, Niagara smoke and CCTV post-process treatment.
- Implemented HLSL noise, Depth Fade, Fresnel, refraction, emissive blending, World Position Offset, material parameter collections, GPU profiling and scalable quality variants.
- Balanced technical art effects against performance by separating expensive translucent effects from cheaper material variants and using profiling to guide visual choices.

AI Stealth Heist Prototype *Unity | C#*
AI and Gameplay Programmer
- Implemented layered guard AI with patrol, suspicious, investigate, search, chase and attack states driven by FSM logic, NavMesh routes, raycast vision and player noise.
- Built multi-sense detection using hearing, smell trails, bait distraction, suspicion levels, global alert sharing and debug visualization to make AI behavior easier to tune.

GrippyGroove *Unity | C#*
Gameplay Programmer
- Developed grappling-hook movement with raycast targeting, hook prediction, line rendering, swing-style motion, controlled release timing, player states and landing recovery.
- Integrated checkpoints, respawn flow, interaction feedback, switches, laser gates, projectile pressure, enemy damage feedback, audio cues and clean team system handoff.
- Separated traversal states so normal movement, grappling, swinging, jumping, falling, landing and recovery did not fight each other during play.

SpiderSquish *Unity | C#*
Gameplay Programmer
- Built a top-down 2D dash-combat project using reusable OOP systems for player movement, dash attacks, enemy behavior, projectile firing, power-ups, inventory, score and UI.
- Implemented health, collision damage, key reveal, door unlock, spawners, audio triggers, score gates and a readable move-dodge-dash-escape gameplay loop.
- Applied encapsulation, inheritance and polymorphism to keep enemies, power-ups, spawners, inventory, score, audio and UI systems easier to extend.

MiltyKitty *Unity | C#*
Solo Developer
- Built a small complete 2D game loop to practice player control, collision setup, enemy and hazard interactions, menu flow, scene changes, score display, health feedback and audio.
- Connected sprites, prefabs, colliders, scripts, UI, basic animation hooks and feedback into a playable loop instead of isolated test scenes.
- Used the project to practice scene management, prefab setup, input debugging, collision checks and small-scope polish from start to finish.

Experience

Develop Conference Volunteer 2024
- Supported event operations, attendee guidance and on-site coordination in a professional games-industry environment while building exposure to developer talks and networking.
- Developed professional communication, quick problem solving and event-flow awareness around studios, speakers and attendees.

Restaurant Supervisor | Blues Kitchen
- Supervise high-volume service, coordinate teams during busy shifts, handle guest issues and maintain standards under pressure.
- Use shift planning, handover, prioritization and calm communication to keep teams aligned during peak service.

Head Waiter / Supervisor | Dishoom
- Led sections, trained staff, managed guest experience and solved service problems during peak trading periods.
- Built leadership habits around coaching, standards, accountability and fast decision making in a high-pressure venue.